

# Boundary Horizons

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Boundary horizons are central to the theory. A horizon is not only an edge. It is a relation between observer, tool, condition, and field of possibilities. What lies beyond the horizon is not necessarily nonexistent; it is only not directly actionable at that moment. This matters in education, law, medicine, physics, AI, art, and governance.

The deep failure is confusing inaccessibility with nonexistence. A learner who does not yet understand is not incapable. Evidence not found does not prove that an event did not occur. Pain not visible in a test does not vanish. A cosmological phenomenon beyond a cosmic event horizon is not false merely because it cannot be seen. Yet beyond the horizon is not a license to invent. A horizon demands double humility: do not erase, and do not fantasize.

Action under a horizon = what can be seen × reliable tool × recognition of limitation × marked uncertainty × possibility of correction / pretended certainty × projection × fear × laziness × authority hiding its conditions. It is a map of responsibility: how to act without pretending that the world ends at the boundary of the instrument.

In science, boundary horizons appear in instruments, energy scales, resolution, observation, and cosmology. In black hole physics, an event horizon is a boundary at which information and action acquire a different meaning for an external observer. In cosmology, a cosmic event horizon marks what can ever influence or be seen. These are physical concepts, not ornaments; their use in the theory must be structural and careful.

In law, the evidence horizon determines responsible judgment. In education, the horizon of understanding determines what distance can be carried. In medicine, the diagnostic horizon determines what is known, suspected, and still needs testing. In AI, the horizon of explanation determines whether a system can be a trustworthy tool or a black box disguised as authority.

Boundary horizons link to the recursive edge because every moved horizon reveals another horizon. There is no final point where all hiddenness disappears. But this is not despair: naming the boundary makes action more moral. A person is not required to know everything; they are required not to lie about what they do not know.

## Source discipline

Working sources and caution: this chapter relies on the theory itself and on background sources checked outside the prompt: UNESCO on artificial intelligence in education, WHO on ethics and governance of AI for health, the NIST AI RMF for AI risk management, CERN/ATLAS on the Standard Model and the Higgs, NASA and ESA/Planck on cosmology, the Stanford Encyclopedia of Philosophy on Bell, Kochen-Specker and contextuality, and Britannica/ISFDB for the literary identification of Harlan Ellison and I Have No Mouth, and I Must Scream. These sources are not treated as proofs of the theory; they are safeguards against empty borrowing of terms.